Franklin Diaz

Lecturer — Information Systems | Principal Security Consultant

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Education

Master of Science in Computer Security
DePaul University, Chicago, IL (2007)

Master of Science in Information Systems
Northwestern University, Evanston, IL (2004)

Bachelor of Science in Computer Science
Roosevelt University, Chicago, IL (2002)

Academic Appointments

Lecturer, Information Systems Discipline — **Sep 2025 – Present**
University of Colorado Denver Business School — Denver, CO

- Teaches **ISMG 2050: Business Problem Solving Tools**, introducing IT infrastructure, data management systems, analytical modeling, and computational business problem solving.

Adjunct Faculty — Information Systems — **2006 – 2011**
McHenry County College — Crystal Lake, IL

- Taught undergraduate courses in Linux OS fundamentals, shell programming, enterprise productivity suites, and Microsoft Windows operating systems.

Professional Experience

Principal Security Consultant / Extended Expertise — **Jan 2020 – Present**
Palo Alto Networks — Remote

- Lead architecture and deployment of Palo Alto Networks security solutions for enterprise clients executing multi-cloud migrations (GCP, OCI, AWS, Azure).
- Architect scalable infrastructure using Infrastructure as Code (IaC), aligning mission-critical environments to rigorous zero-trust reference architectures.
- Serve as technical advisor identifying policy and architectural gaps, delivering remediation roadmaps, and advising executive stakeholders post-incident.

Security Engineer (Detection & Response Engineering) — **Oct 2014 – Jan 2020**
Salesforce — Bellevue, WA

- Built high-throughput security telemetry collection systems processing **over 1 billion security events per day** across global enterprise infrastructure.
- Engineered and maintained analytical data pipelines in a FedRAMP High GovCloud environment on AWS.
- Developed distributed packet capture infrastructure, CI/CD automation, and integrated regression test suites across multi-repository toolsets.

Principal Research Software Engineer — **Aug 2012 – Jan 2014**
Role: Security Engineer & Test Architect | *Indiana University (SDN Lab) — Bloomington, IN*

- Directed testing strategies and test case engineering for Software Defined Networking (OpenFlow) infrastructure.
- Developed web-based telemetry and management tools for the GENI/GEMINI national experimental network testbed.
- Assisted undergraduate computer networking courses: led discussions, developed coursework, and proctored examinations.

Research & Development Engineer **Apr 2011 – Jul 2012**
Location: Espoo, Finland & Arlington Heights, IL | *Nokia Siemens Networks*

- Conducted cellular base station product development, firmware integration, and multi-subsystem verification for CDMA telecom systems.

Senior Software Engineer **Jan 1998 – Mar 2011**
Cellular Networks & Space Sector — 13-Year Core Engineering Tenure | *Motorola Solutions*

- Engineered CDMA cellular base station infrastructure and embedded subsystems using C and C++.
- Partnered directly with premier carrier account teams including Sprint, ALLTEL, Lucent, Pelephone, and NTT on systems integration and network testing.
- Led cross-functional system integration efforts and acceptance testing across distributed product feature teams.

Projects & Open Research

- **Cloud Native Security Lab** (Zenodo Record 5918861): Complete open-source cloud-native security research lab deployed in Google Cloud Platform.
- **project emerald**: Minecraft schematic generation utilizing local AI Edge TPU hardware acceleration.
- **Peirates**: Contributor to open-source Kubernetes penetration testing and attack-surface evaluation tool.

Publications, Talks & Community Service

- **Conference Security Track Volunteer:** BSidesLV.
- **Technical Author & Researcher:** Decentralized protocols, autonomous agent runtime security, and DevSecOps at bitsmasher.net/research.

Technical Skills & Competencies

Languages:	Python, SQL, Go, JavaScript, Bash, C/C++, Java, PERL
Frameworks & Libraries:	Pydantic, FastAPI, Django, Leaflet, LINQ
Cloud & Platforms:	AWS, GCP, Azure, OCI
Tools & Infrastructure:	Git, JIRA, Kafka, Redis, Docker, Kubernetes, Terraform, Airflow, Jenkins, LaTeX
Concepts & Practices:	DevSecOps, Infrastructure as Code, REST/XML APIs, Object-Oriented Design, TDD, Multi-Cloud Architecture